Satirtha Paul Shyam

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Education

Bangladesh University of Engineering and Technology (BUET) — *Dhaka, Bangladesh*
M.Sc. in Electrical and Electronic Engineering (Major: Communication and Signal Processing) — Apr. 2023 - Jun. 2025
CGPA: 3.50 (on a scale of 4.00)

M.Sc. Thesis
Probabilistic Deep Learning Framework Based on Normalizing Flow Algorithm for Ambiguous Medical Image Segmentation — Apr. 2024 – Jun. 2025
Resulted in 1 Q1 journal publication and 1 Q1 journal manuscript under review.

Military Institute of Science and Technology — *Mirpur Cantonment, Dhaka, Bangladesh*
B.Sc. in EECE (Electrical, Electronic and Communication Engineering) — Feb. 2019 - Mar. 2023
CGPA: 3.72 (on a scale of 4.00)

B.Sc. Thesis
Segmentation of Noisy and Intensity Inhomogeneous Images Using Regional Scalable Fitting and Difference of Gaussian — Mar. 2022 - Feb. 2023
Supervisor: Asst. Prof. Chowdhury Mohammad Abid Rahman

Notre Dame College — *Motijheel, Dhaka, Bangladesh*
Higher Secondary in Science — May 2016 - Jul. 2018
GPA: 5.00 (on a scale of 5.00)

Comilla Modern High School — *Kandirpur, Cumilla, Bangladesh*
Secondary in Science — Feb. 2012 - Mar. 2016
GPA: 5.00 (on a scale of 5.00)

Technical Skills

Programming	Matlab, C, C++, Python, Dart
Professional Softwares	Simulink, Proteus, Cadence, PSpice, AutoCAD, OrCAD, DSCH2, Microwind, ETAP
Drawing & Typesetting	Illustrator, Office, LaTeX
3d simulation	Blender, Unity Engine
Languages	Bangla(Native), English(speaking and writing), Hindi(speaking)

Publications

Journal Articles (2)

Satirtha Paul Shyam, Shaikh Anowarul Fattah, Mohammad Saquib, “A Multi-Level Probabilistic Deep Learning Network Augmented With Normalizing Flow for Ambiguous Medical Image Segmentation,” *IEEE Access*, vol. 14, pp. 4063–4079, 2025.

Salma Sultana Mim, **Satirtha Paul Shyam**, Chowdhury M. Abid Rahman, “A Median Based Signed Pressure Force Driven Active Contour Model with Improved Pre-Processing for Image Segmentation,” *SSRN Preprint*, 2025.

Conference Papers (12)

Satirtha Paul Shyam, CMA Rahman, H. Rashid, “A Faithful DoG is All You Need,” *25th International Conference on Computer and Information Technology (ICCIT)*, IEEE, pp. 751–756, 2022.

Satirtha Paul Shyam, CMA Rahman, PK Partho, RK Bhuiyan, "Edge and Region Synergy Driven Active Contour Models for Natural Image Segmentation," *5th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, IEEE, pp. 1–6, 2023.

CMA Rahman, **Satirtha Paul Shyam**, PK Partho, Wasi Mashrur Shian, "Noise Optimized Order-Statistic Energy Pre-fitted Active Contour for Accurate Image Segmentation," *33rd International Conference on Computer Theory and Applications (ICCTA)*, IEEE, pp. 89–94, 2023.

Satirtha Paul Shyam, CMA Rahman, PK Partho, RK Bhuiyan, "Non-linear Statistical Filter Fusions for Optimal Intensity Fitting in Active Contour-Based Image Segmentation," *6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*, IEEE, pp. 190–195, 2024.

Salma S. Mim, **Satirtha Paul Shyam**, CMA Rahman, S. Akter, J. Fardous, M.A. Samad, "A Gaussian-Extremum Image Pre-fitted Active Contour for Accurate and Faster Image Segmentation," *6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*, IEEE, pp. 687–692, 2024.

CMA Rahman, Rahat K. Bhuiyan, **Satirtha Paul Shyam**, Rumana Subnom, Adib Bin Rashid, "Attention Enabled MultiResUNet for Bio-Medical Image Segmentation," *6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*, IEEE, pp. 622–627, 2024.

Satirtha Paul Shyam, Rahat K. Bhuiyan, Salma S. Mim, CMA Rahman, "An Advanced Cascaded UNet Architecture with Dual Attention for Bio-Medical Image Segmentation," *IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE)*, IEEE, pp. 1–6, 2024.

MSR Chowdhury, PK Partho, CMA Rahman, **Satirtha Paul Shyam**, SS Mim, RK Bhuiyan, "Enhancing Coronary Disease Prediction Using Stacking and Voting-Based Ensemble Machine Learning Models," *27th International Conference on Computer and Information Technology (ICCIT)*, IEEE, pp. 2829–2834, 2024.

Monjila Afrin Dorothi, Shaharior Anik, Sourav Chandra Das, **Satirtha Paul Shyam**, Mst. Umme Habiba, "Circular PCF SPR Sensor Design with External Sensing and Extended RI Range," *7th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, IEEE, pp. 1–6, 2025.

Md Nayeem Sharker, Lusain Nima, Mohibul Islam Sazib, Md Badrul Ahsan Tamal, Saiful Islam, **Satirtha Paul Shyam**, "Stability-Oriented Bi-Directional EV Charger with PLL-Based Synchronization and PI Control," *7th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, IEEE, pp. 1–6, 2025.

Sourav Chandra Das, Mst. Umme Habiba, **Satirtha Paul Shyam**, Nipa Mridha, Farjana Yasmin, Nafisa Anjum Prapti, Wasi Mashrur Shian, Md Nayeem Sharker, "A Sustainable IoT-Based Atmospheric Water Generator for Portable Clean Water Harvesting," *7th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, IEEE, pp. 1–5, 2025.

Satirtha Paul Shyam, Proshit Kumar Partho, Md Shahnewaz, Md Nayeem Sharker, Salma Sultana Mim, "Advancing Diabetes Diagnosis via Multi Stage Ensemble Learning and Deep Neural Architectures with Enhanced Data Preprocessing," *2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, IEEE, pp. 1–6, 2026.

Teaching Experience

Department of Electrical and Electronic Engineering — *Green University of Bangladesh, Dhaka, Bangladesh*

Lecturer — February 2024 – Present

- Teaching undergraduate courses in Electrical and Electronic Engineering.
- Courses taught:
  - EEE 335 – Digital Signal Processing
  - EEE 309 – Solid State Devices
  - EEE 205 – Engineering Electromagnetics
  - EEE 317 – Microprocessor and Interfacing
  - EEE 307 – Communication Theory
  - MATH 301 – Numerical Methods
  - EEE 101 – Introduction to Electrical Engineering
- Preparing lecture materials, assignments, quizzes, examinations, and laboratory sessions.
- Mentoring undergraduate students and supervising academic projects.

Field of Interest

1. Digital Signal Processing
2. Image Processing

3.. Computer Vision

Projects

1. Build an IOT-based automated plant water system using Arduino
2. Designed and simulated a simple energy meter circuit using Proteus software
3. Build an op-amp using Cadence
4. Build a cinematic animation and a photorealistic animation as a personal project by blender

Awards and Honors

2020	**Award**: "Dean's List position for academic excellence"	*Level: 2*
2022	**Award**: "Dean's List position for academic excellence"	*Level: 4*
2017	**Achievement**: "secured 18th position in 7th Physics Olympiad"	*Category:C*
2016	**Achievement**: "secured 23rd position in 6th Physics Olympiad"	*Category:B*

Coursework Information

1. Digital Signal Processing
2. AI and Machine Learning
3. VLSI
4. Continuous Signal and Linear Systems
5. Electrical Machines (AC and DC)

Industrial Training

Practical Training for 1 month in the following Industries and Institutions as industrial trainee:

1. Fair Electronics Factory-Samsung
2. Power Grid Company of Bangladesh(PGCB)
3. Energypac Engineering
4. Bangladesh Betar
5. Teletalk Bangladesh Ltd.
6. Bangladesh Telecommunication Regulatory Commission(BTRC)

Certificates

1. Programming for Everybody (Getting Started with Python) an online non-credit course authorized by University of Michigan and offered through Coursera
2. Using Python to Access Web Data an online non-credit course authorized by University of Michigan and offered through Coursera
3. Introduction to the Internet of Things and Embedded Systems an online non-credit course authorized by University of California, Irvine and offered through Coursera
4. The Arduino Platform and C Programming, an online non-credit course authorized by University of California, Irvine and offered through Coursera
5. Electrodynamics: An Introduction an online non-credit course authorized by Korea Advanced Institute of Science and Technology(KAIST) and offered through Coursera
6. Introduction to Electronics an online non-credit course authorized by Georgia Institute of Technology and offered through Coursera
7. Introduction to Flutter development using Dart

References

Dr. Shaikh Anowarul Fattah
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Bangladesh University of Engineering and Technology (BUET)
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Chowdhury Mohammad Abid Rahman
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Military Institute of Science and Technology, Dhaka, Bangladesh
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